

PROLOGUE

INTRODUCTION TO COMPUTER SCIENCE · WEEK 2 · STATION I

Size

Bits, orders of magnitude, state spaces — and why testing shows presence, not absence.

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One line, in a formal language.

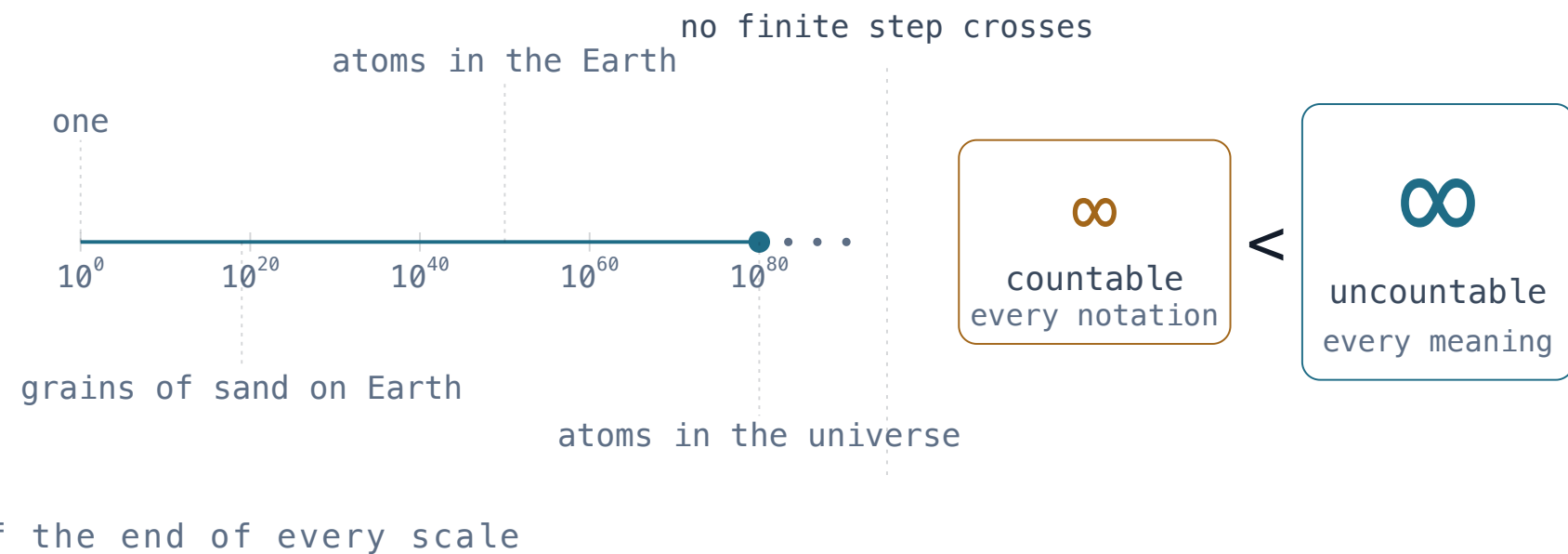
```
$ ./selfie
./selfie { -c { source } | -o binary | [ -s | -S ] assembly | -l binary } [ ( -m | -d | -r | -y ) 0-40
```

This is selfie's synopsis: a regular expression that says exactly how selfie may be invoked. Braces mean *any number of*, brackets mean *optional*, the bar means *or*. We learn to read it properly in week 4.

For today: `-c selfie.c` compiles selfie's own source. Watch what it prints about itself.

```
$ ./selfie -c selfie.c
./selfie: selfie compiling selfie.c to 64-bit RISC-U with 64-bit starc
./selfie: 365784 characters read in 12394 lines and 1741 comments
./selfie: with 219888(60.11%) characters in 51329 actual symbols
./selfie: 491 global variables, 661 procedures, 512 string literals
./selfie: 188392 bytes generated with 43492 instructions and 14424 bytes of data
```

One line carries the whole class: small, vast, then two sizes of endless.



Left. From one thing to every atom there is — the whole physical world in eighty steps, each one ten times the last.

The wall. No number of steps crosses it. Infinity is not the far end of the line; it is what the line never reaches.

Right. Past the wall, endlessness comes in sizes: one counts every notation, the other every meaning — and is strictly bigger.

ANCHORS

A million, a billion, a trillion. Three words
one syllable apart — **three different worlds.**

10^6 s

A MILLION SECONDS · $11\frac{1}{2}$
DAYS

10^9 s

A BILLION SECONDS · 32
YEARS

10^{12} s

A TRILLION SECONDS ·
32,000 YEARS

A million seconds ago you had not yet packed
for today. A billion seconds ago you were a
child, or not yet born. A trillion seconds ago
somebody was painting the walls of the Chauvet
cave.

THE HABIT

Never accept a number you cannot picture. Find the
anchor — a fortnight, a career, an ice age — or admit
you are repeating the number, not reading it.

UNITS

Kilo, mega, giga, tera: the same three zeros, again and again.

PREFIX	FACTOR	BYTES · HOW MUCH IT HOLDS	HERTZ · HOW OFTEN IT HAPPENS
kilo	10^3	KB – a long paragraph	kHz – the top of human hearing, 20 kHz
mega	10^6	MB – a novel	MHz – the first IBM PC, 4.77 MHz, 1981
giga	10^9	GB – a thousand novels	GHz – this laptop; Wi-Fi at 2.4
tera	10^{12}	TB – a million novels	THz – infrared light

A byte is eight bits — one keystroke. A hertz is one beat per second. The left column counts things, the right counts events, and the prefixes do not care which.

HOLD A NANOSECOND

At 1 GHz a beat lasts a billionth of a second, and light gets 30 cm in that time. Grace Hopper handed those 30 cm out as lengths of wire, so a room could feel why a signal cannot cross a desk and return inside one beat.

TWO KILOS

Kilo means a thousand. In memory it means 1,024. Both are in daily use.

	BASE 10	BASE 2	GAP
kilo	KB = 10^3	KiB = 2^{10} = 1,024	+2.4%
mega	MB = 10^6	MiB = 2^{20}	+4.9%
giga	GB = 10^9	GiB = 2^{30}	+7.4%
tera	TB = 10^{12}	TiB = 2^{40}	+10.0%

Memory is addressed with bits, so its natural steps are powers of two — and 2^{10} lands so close to 1,000 that the prefix was simply borrowed. Memory comes in powers of two; disks and networks in powers of ten. A "1 TB" drive holds 10^{12} bytes and shows up as 931 GiB.

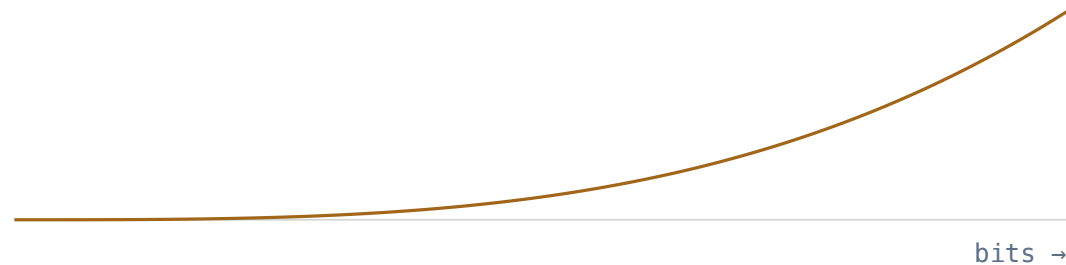
NOTICE WHAT JUST HAPPENED

One notation, two meanings, and nothing in the notation to tell you which you were handed. Keep hold of that. The rest of the semester is about that gap.

A bit is *one distinction*: this, not that.



$$2^{16} = 65,536 \text{ states}$$



$$\text{state space} = 2^n$$

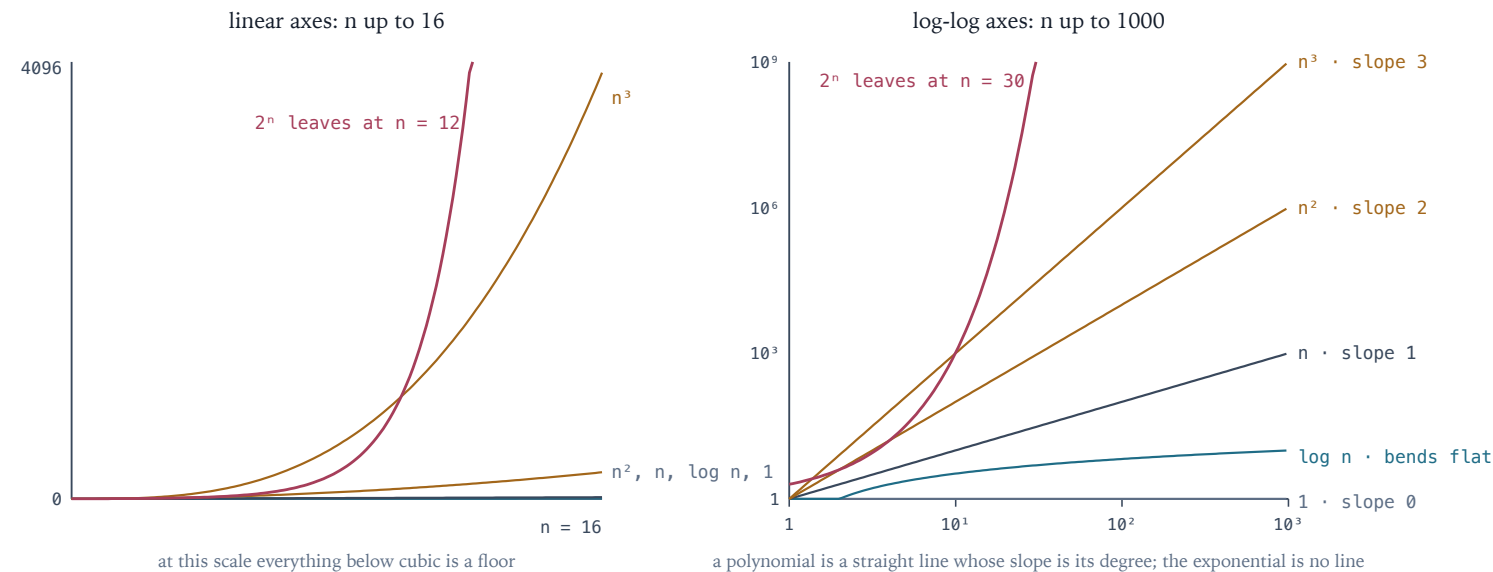
Add a bit and you don't *add* states. You *double* them.

Ten bits: a thousand states. Twenty: a million. Thirty: a billion.

Doubling is the most underestimated operation in human reasoning — and the entire engine of computing.

How many bits does it take to have more states than the universe has atoms?

Constant, logarithmic, linear, quadratic, cubic — and then something that is **not on the same list**.



the same six functions, twice

Left, ordinary axes: by $n = 16$ the cube has reached 4,096 and everything below it is a floor. The exponential left the page at $n = 12$.

Right, both axes logarithmic, which is the only way to see them together. Every polynomial becomes a straight line and its slope is its degree: 0, 1, 2, 3. The logarithm bends flat. The exponential bends *up*, and leaves at $n = 30$, when the cube is still at 27,000.

Polynomials differ from each other by a slope. The exponential differs from all of them by kind.

Multiply the input by ten. A polynomial multiplies by a constant. Add one to the input. An exponential multiplies by a constant.

	N = 10	N = 11	N = 100	N = 1,000
1	1	1	1	1
$\log_2 n$	3.3	3.5	6.6	10
n	10	11	100	1,000
n^2	100	121	10,000	10^6
n^3	1,000	1,331	10^6	10^9
2^n	1,024	2,048	10^{30}	10^{301}

Times ten. Read the columns $n = 100$ and $n = 1,000$. Linear: $\times 10$. Quadratic: $\times 100$. Cubic: $\times 1,000$. A constant factor each time, fixed by the degree; that is what *polynomial* means. The exponential goes from 10^{30} to 10^{301} , and there is no factor.

Plus one. Read $n = 10$ and $n = 11$. Every polynomial moves by a little: 10 percent, 21 percent, 33 percent. The exponential doubles. That is the test: if adding one to the input multiplies the output, you are looking at an exponential. Every bit added to a memory is that plus one.

And the logarithm is the exponential read backwards: multiply the input by ten and it goes up by a constant, 3.3. Ten times the states, three more bits. That is why 266 bits are enough for the universe.

SCALE CHECK

34 bytes of memory has more states than the observable universe has atoms.

 10^{80}

ATOMS IN THE OBSERVABLE UNIVERSE

 2^{266} STATES OF 266 BITS $\approx 1.2 \times 10^{80}$ 37 bytes to beat every photon, about 10^{89} .

A phone with 8 GB of memory has $2^{68,719,476,736}$ states.

Written out in decimal, that number has about 20.7 billion digits. At 3,000 digits a page: 6.9 million pages. Bound into 500-page books: 13,800 volumes — some 400 metres of shelf.

That shelf does not hold the phone's states. It holds the *number* of them, written down once.

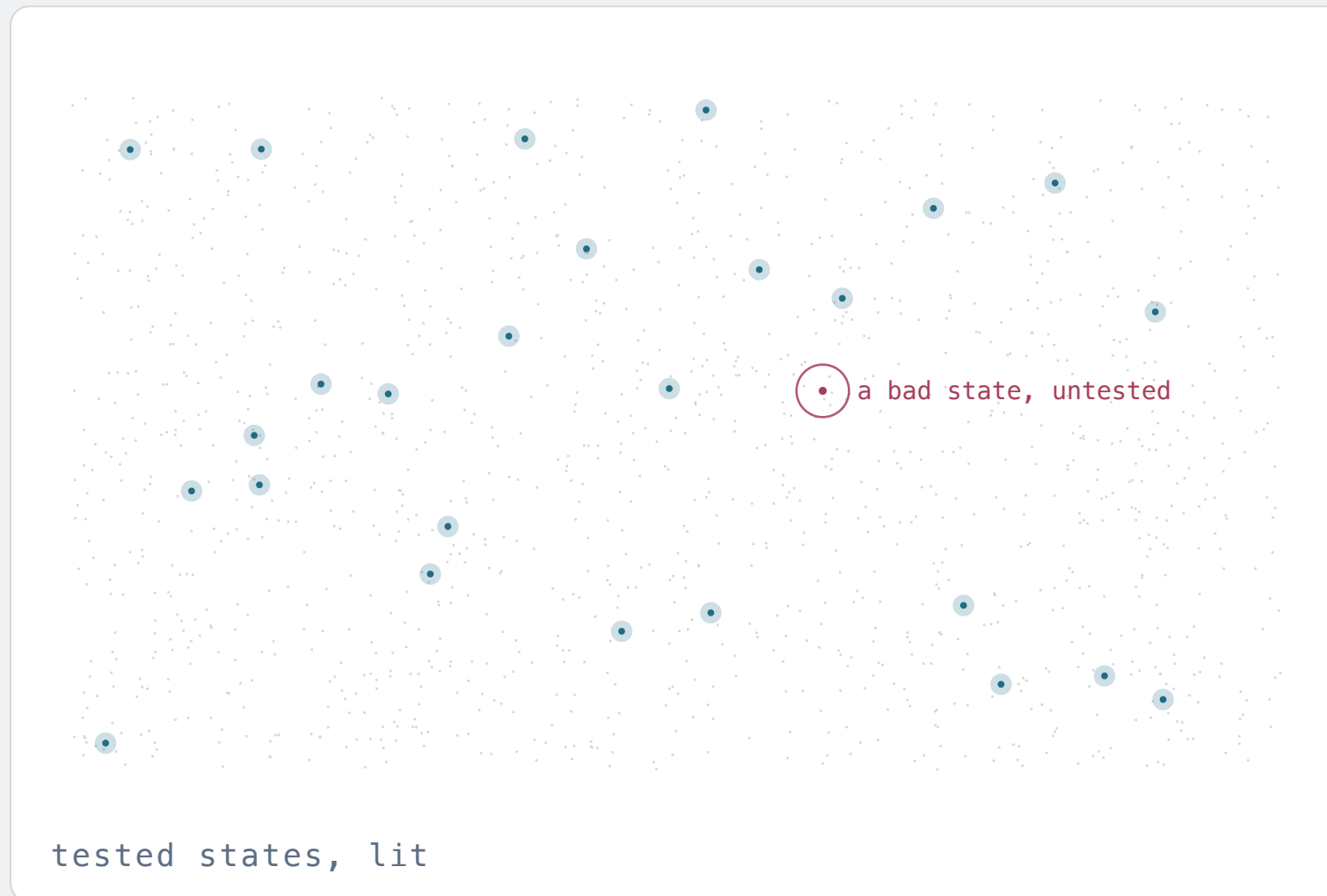
Every app you will ever install, every photo you will ever take, and every bug you will ever hit is one point in this space.

SELFIE'S MACHINE

RISC-U has 4 GB of byte-addressed memory: 2^{35} bits, so $2^{34,359,738,368}$ states — a number with 10.3 billion decimal digits. Selfie, the program that runs on it, is 12,394 lines. One person can read the program to the end. Nobody can inspect the machine.

$2^{34359738368} = 23873117371686870504051737186753437$
 903168979747538053465075696759119864115092050339869
 941054292537183290022099175144573637279641592487667
 345504934445665896266534390508060257364588787358720
 564536368615392783093371704721401685574994838137187
 636205568269232431289180378602417641004581755214355
 054030083542708703703070552420072448203831600008200

In a space that big, good states and bad states **look alike**.



Test a billion states per second, starting at the Big Bang, and by now you would have checked about 2^{88} of them. Out of $2^{68,719,476,736}$.

The fraction is not small. It is indistinguishable from zero.

Testing shows the presence, not the absence of bugs.

EDSGER W. DIJKSTRA, 1969

The vastness is not the problem. It is the **inventory**.

WHAT IT FORBIDS

Certainty by inspection. You will never check your system exhaustively — not with more testers, not with faster machines, not ever.

WHAT IT OPENS

Every symphony not yet written, every protein not yet folded, every proof not yet found, every program not yet imagined — all of them are already in there, waiting to be addressed. The same enormity that hides your bugs is the reason novelty is inexhaustible.

First appearance of the device that every limit in this class gets: what it forbids, and what it opens. A space small enough to audit would be a space too small to be interesting.

Borges built this space in 1941 and called it a **library**.

The Library of Babel holds every possible book: every truth, every refutation — and overwhelmingly, gibberish. Its inhabitants go mad: containing every truth is worthless without a way of finding one.

Modern echo: a large language model already "contains" astonishing amounts of text. Containment was never the hard part.

BORGES · 1941

410 pages, 40 lines, 80 characters, 25 symbols. The shelves hold $25^{1,312,000}$ books $\approx 10^{1,834,097}$ volumes.

THE SHIFT

From storage to search: from having the answer somewhere to knowing it when you see it.

A machine that counts through all its states would, eventually, write the Mona Lisa. It would not **notice**.

Write a program that treats all of memory as one integer and increments it. It enumerates every state the machine can be in. It will have turned to dust before getting anywhere — but suppose it ran.

Once in a while, an enormous while, the state would encode Beethoven's Fifth, or the human genome. Without any intelligence whatsoever. And the machine could not *select* those states over the others.

Evolution is that program with a selector attached: enumerate DNA by mutation, select by reproductive success. A physical process bound by physics, modelled by a computer bound by logic.

QUANTITIES MATTER

Lots of time, or mutate fast. Lots of space, or be tiny. Lots of energy, or be efficient. A few more zeroes in a number make all the difference — in life and in computing.

Recommended exercises.

- 1 Read the Size chapter up to *Numbers*, and Borges's *Library of Babel*.
- 2 Rerun `./selfie -c selfie.c`. Where on the map do 365,784 characters sit? And 43,492 instructions? And the machine's state count?
- 3 For each of n , n^2 , n^3 and 2^n : what happens to the value when n goes from 10 to 11, and when it goes from 10 to 100? Which of the four is the odd one out, and in which of the two questions?
- 4 How many bits do you need for a thousand states? A million? Every person alive? Every atom in the Earth (about 10^{50})?
- 5 A "512 GB" phone: how many GiB does it report, and why?
- 6 Find three anchors of your own for a million, a billion, and a trillion — of anything.

NEXT WEEK

Everything is bits: numbers, negative numbers, overflow, characters, text, files, images, code. The same 85 in five different notations, and what the machine makes of each.

Listen, then read.

LISTEN

[Beethoven · Symphony No. 5](#) — Carlos Kleiber, Vienna Philharmonic, 1974. Four notes, and the whole first movement — the whole symphony — is built from them. Small to vast in one line. Then watch Leonard Bernstein explain the first movement [on YouTube](#) on the 1954 Omnibus broadcast: how Beethoven found the notation he needed, one rejected sketch at a time.

READ

[One Two Three... Infinity](#) — Gamow, 1947: from counting to the sizes of endless, by a physicist who could draw. And, more technical, [Mathematics and Computer Science: Coping with Finiteness](#) — Knuth, 1976: numbers so large that finite stops meaning small.